Week #6: Least Squares and QR Decomposition

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Overview

Linear least squares problems arise naturally when we have more equations than unknowns, leading to overdetermined systems that generally have no exact solution. This week we develop the mathematical theory for finding the "best" approximate solution in the least squares sense, exploring both the normal equations approach and the more numerically stable QR decomposition method.

The least squares problem represents a fundamental paradigm in numerical analysis: when exact solutions don't exist, we seek optimal approximate solutions according to well-defined criteria. This leads us to explore orthogonal transformations, particularly the QR decomposition, which provides both theoretical insights and practical algorithms for solving overdetermined systems.

Our analysis emphasizes the geometric interpretation of least squares as orthogonal projection onto subspaces, the role of orthogonal matrices in preserving numerical properties, and the superior stability characteristics of QR-based methods compared to normal equations. These concepts form the foundation for understanding more advanced topics in numerical linear algebra and optimization.

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1 The Linear Least Squares Problem

1.1 Problem Formulation

Definition. Linear Least Squares Problem

Given $A \in \mathbb{R}^{m\times n}$ with $m \geq n$ and $b \in \mathbb{R}^m$, find $x \in \mathbb{R}^n$ that minimizes:

$$\|Ax - b\|_2^2 = \sum_{i=1}^{m} (a_i^T x - b_i)^2$$

where a_i^T is the i-th row of A.

The geometric interpretation reveals that we seek the point Ax in the column space of A that is closest to b in the Euclidean norm. This optimization problem always has a solution, and the solution is unique when A has full column rank.

Theorem 1.1 (Existence and Uniqueness of Least Squares Solutions)**.** *The linear least squares problem* $\min_x \|Ax - b\|_2$ *always has a solution. The solution is unique if and only if A has full column rank.*

Proof. Existence follows from the compactness of the feasible set and continuity of the objective function. For uniqueness, if A has full column rank, then $A^T A$ is positive definite, making the normal equations have a unique solution. □

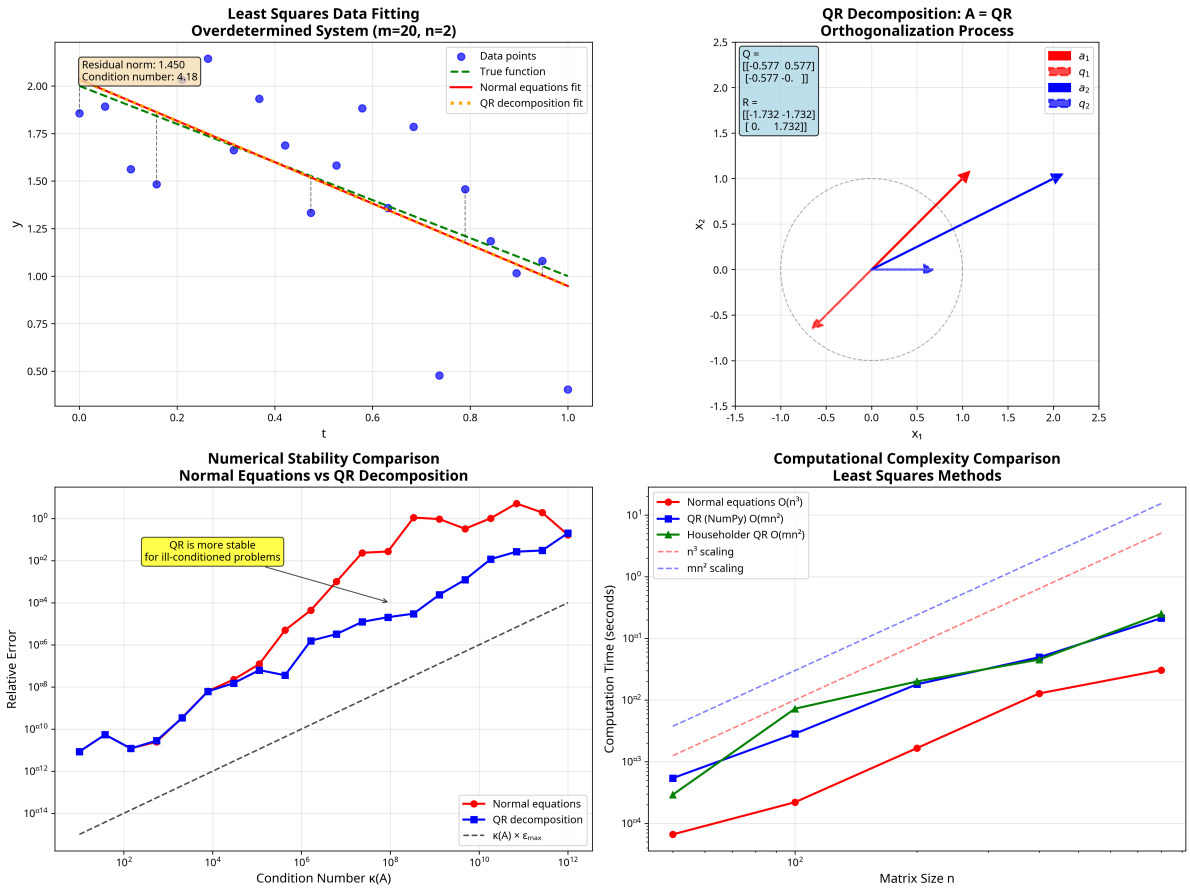

Figure 1: Comprehensive least squares analysis showing geometric interpretation, stability comparison between normal equations and QR decomposition, computational complexity analysis, and conditioning effects. The top-left panel demonstrates data fitting with residual visualization, while other panels show numerical stability, complexity scaling, and method comparisons.

1.2 Geometric Interpretation

The least squares solution has a beautiful geometric interpretation in terms of orthogonal projections.

Theorem 1.2 (Orthogonal Projection Characterization)**.** *The least squares solution x^* satisfies $Ax^* = P_A b$, where $P_A = A(A^T A)^{-1} A^T$ is the orthogonal projection onto the column space of A.*

The residual $r^* = b - Ax^*$ is orthogonal to the column space of A, which gives us the fundamental relationship:

$$A^T r^* = A^T (b - Ax^*) = 0$$

This orthogonality condition is the key to deriving both the normal equations and understanding why QR decomposition provides a stable solution method.

Intuition: Geometric Understanding

Think of the least squares problem as finding the point in a subspace (column space of A) that is closest to a given point (b). The closest point is found by dropping a perpendicular from b to the subspace. This perpendicular direction is exactly the residual vector, and the requirement that it be orthogonal to the subspace gives us our optimality conditions.

2 Normal Equations Approach

2.1 Derivation and Properties

The classical approach to solving least squares problems uses the normal equations, derived from the optimality condition.

Theorem 2.1 (Normal Equations)**.** *The vector x^* minimizes $\|Ax - b\|_2$ if and only if it satisfies the normal equations:*

$$A^T Ax^* = A^T b$$

Proof. The objective function $f(x) = \|Ax - b\|_2^2 = (Ax - b)^T(Ax - b)$ has gradient $\nabla f(x) = 2A^T(Ax - b)$. Setting this to zero yields the normal equations. □

2.2 Computational Algorithm

The normal equations approach involves the following steps:

1. Form the matrix $A^T A$ and vector $A^T b$
2. Solve the symmetric positive definite system $(A^T A)x = A^T b$
3. This can be done efficiently using Cholesky decomposition when A has full column rank

The computational cost is approximately mn^2 operations to form $A^T A$ and $A^T b$, plus $\frac{n^3}{3}$ operations for the Cholesky decomposition and solution.

2.3 Numerical Stability Issues

The major drawback of the normal equations approach is numerical instability for ill-conditioned matrices.

Theorem 2.2 (Condition Number Squaring)**.** *If $\kappa(A) = \|A\|_2\|A^+\|_2$ is the condition number of A, then:*

$$\kappa(A^T A) = \kappa(A)^2$$

This means that forming $A^T A$ squares the condition number, potentially leading to severe loss of accuracy. For example, if $\kappa(A) = 10^6$, then $\kappa(A^T A) = 10^{12}$, which approaches the limit of double precision arithmetic.

3 QR Decomposition

3.1 Definition and Existence

Definition. QR Decomposition

For $A \in \mathbb{R}^{m\times n}$ with $m \geq n$, a QR decomposition is a factorization:

$$A = QR$$

where $Q \in \mathbb{R}^{m\times m}$ is orthogonal ($Q^T Q = I$) and $R \in \mathbb{R}^{m\times n}$ is upper triangular.

For computational purposes, we often use the reduced QR decomposition:

$$A = Q_1 R_1$$

where $Q_1 \in \mathbb{R}^{m\times n}$ has orthonormal columns and $R_1 \in \mathbb{R}^{n\times n}$ is upper triangular.

Theorem 3.1 (Existence of QR Decomposition)**.** *Every matrix $A \in \mathbb{R}^{m\times n}$ has a QR decomposition. If A has full column rank, then R can be chosen to have positive diagonal entries, making the decomposition unique.*

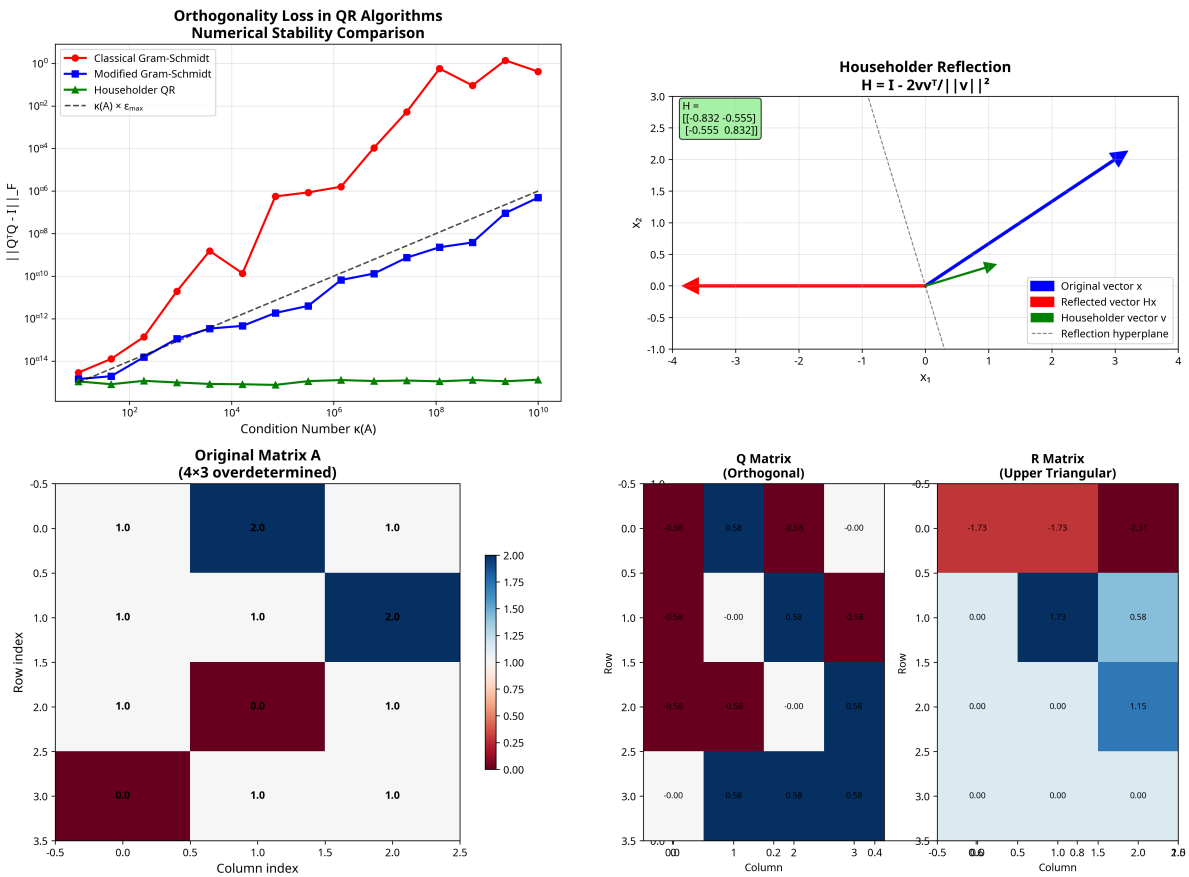

Figure 2: QR decomposition algorithms comparison showing orthogonality loss analysis, Householder reflection visualization, and step-by-step matrix decomposition. The top-left panel demonstrates the superior numerical stability of Householder QR over Gram-Schmidt variants, while the top-right shows the geometric interpretation of Householder reflections.

3.2 Gram-Schmidt Process

The classical Gram-Schmidt process provides a constructive proof of QR decomposition existence.

Definition. Classical Gram-Schmidt

Given linearly independent vectors $a_1, \ldots, a_n$, the Gram-Schmidt process constructs orthonormal vectors $q_1, \ldots, q_n$:

$$v_1 = a_1, \quad q_1 = \frac{v_1}{\|v_1\|} \tag{1}$$

$$v_k = a_k - \sum_{j=1}^{k-1} (q_j^T a_k) q_j, \quad q_k = \frac{v_k}{\|v_k\|} \tag{2}$$

This process is numerically unstable due to loss of orthogonality in finite precision arithmetic.

Definition. Modified Gram-Schmidt

The modified Gram-Schmidt process improves numerical stability by orthogonalizing against each q_j immediately after computing it:

$$v_1^{(1)} = a_1, \quad q_1 = \frac{v_1^{(1)}}{\|v_1^{(1)}\|} \tag{3}$$

$$v_k^{(k)} = a_k \tag{4}$$

$$v_k^{(j)} = v_k^{(j-1)} - (q_{j-1}^T v_k^{(j-1)}) q_{j-1}, \quad j = 2, \ldots, k \tag{5}$$

$$q_k = \frac{v_k^{(k)}}{\|v_k^{(k)}\|} \tag{6}$$

3.3 Householder Reflections

Householder reflections provide the most numerically stable method for computing QR decompositions.

Definition. Householder Reflection

For a nonzero vector $v \in \mathbb{R}^m$, the Householder reflection is:

$$H = I - 2\frac{vv^T}{\|v\|^2}$$

This matrix is symmetric, orthogonal, and reflects vectors across the hyperplane orthogonal to v.

Theorem 3.2 (Householder QR Algorithm)**.** *For any $A \in \mathbb{R}^{m\times n}$, there exist Householder matrices $H_1, \ldots, H_n$ such that:*

$$H_n \cdots H_1 A = R$$

where R is upper triangular. The QR decomposition is $A = QR$ with $Q = H_1 \cdots H_n$.

The Householder vector for eliminating entries below the diagonal in column k is chosen as:

$$v_k = x_k + \operatorname{sign}(x_{k,k})\|x_k\|e_1$$

where x_k is the k-th column of the current matrix.

4 Solving Least Squares via QR

4.1 QR-Based Solution

Using the QR decomposition, the least squares problem becomes much simpler to solve.

Theorem 4.1 (Least Squares Solution via QR)**.** *If $A = QR$ where Q is orthogonal and R is upper triangular, then the least squares solution is:*

$$x^* = R^{-1}Q^T b$$

assuming R is invertible (i.e., A has full column rank).

Proof. Since Q is orthogonal, $\|Ax - b\|_2 = \|QRx - b\|_2 = \|Rx - Q^T b\|_2$. The minimum occurs when $Rx^* = Q^T b$. □

For the reduced QR decomposition $A = Q_1 R_1$:

$$x^* = R_1^{-1} Q_1^T b$$

4.2 Computational Algorithm

The complete algorithm for solving least squares via QR decomposition:

1. Compute QR decomposition: $A = QR$ (using Householder reflections)
2. Compute $c = Q^T b$
3. Solve upper triangular system: $Rx = c$

The computational cost is approximately $2mn^2 - \frac{2n^3}{3}$ operations for the QR decomposition, plus $O(n^2)$ for the triangular solve.

Listing 1: QR-based Least Squares Solution

```python
import numpy as np
from scipy.linalg import qr, solve_triangular

def qr_least_squares(A, b):
    """Solve least squares problem using QR decomposition"""
    Q, R = qr(A, mode='economic')  # Reduced QR
    c = Q.T @ b
    x = solve_triangular(R, c)
    return x

# Example usage
A = np.random.randn(20, 10)  # Overdetermined system
x_true = np.random.randn(10)
b = A @ x_true + 0.01 * np.random.randn(20)  # Add noise

x_solution = qr_least_squares(A, b)
residual_norm = np.linalg.norm(A @ x_solution - b)
print(f"Residual norm: {residual_norm:.6f}")
```

5 Numerical Stability Analysis

5.1 Condition Numbers and Error Analysis

The conditioning of least squares problems depends on the matrix A.

Definition. Condition Number for Least Squares

For the least squares problem $\min_x \|Ax - b\|_2$, the condition number is:

$$\kappa(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$$

where $\sigma_{\max}$ and $\sigma_{\min}$ are the largest and smallest singular values of A.

Theorem 5.1 (Perturbation Analysis)**.** *For small perturbations ΔA and Δb, the relative error in the least squares solution satisfies:*

$$\frac{\|\Delta x\|}{\|x\|} \leq \kappa(A) \left(\frac{\|\Delta A\|}{\|A\|} + \frac{\|\Delta b\|}{\|b\|} \right) + O(\epsilon^2)$$

5.2 Stability Comparison

Theorem 5.2 (Stability of Normal Equations vs. QR).

- ***Normal equations:*** *The computed solution satisfies* $(A^T A + \Delta(A^T A))\tilde{x} = A^T b$ *where* $\|\Delta(A^T A)\| \leq \gamma_n \|A^T A\|$ *with* $\gamma_n = O(n\epsilon_{mach})$.
- ***QR decomposition:*** *The computed solution satisfies* $(A + \Delta A)\tilde{x} = b$ *where* $\|\Delta A\| \leq \gamma_n \|A\|$.

The key insight is that the condition number of $A^T A$ is the square of the condition number of A, making normal equations potentially much less stable.

Intuition: Why QR is More Stable

The normal equations square the condition number because we compute $A^T A$. QR decomposition works directly with A, preserving its conditioning. Orthogonal transformations (like those in QR) preserve norms and angles, leading to better numerical behavior.

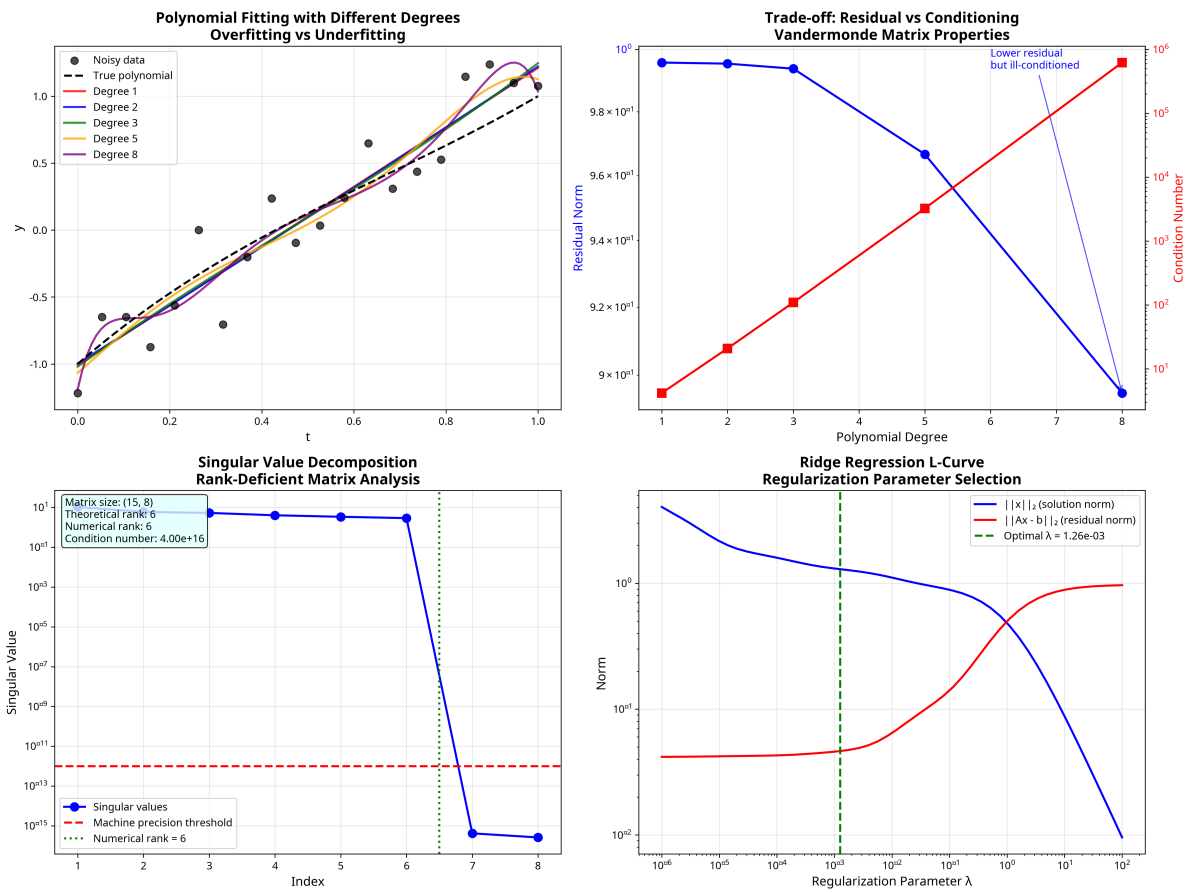

Figure 3: Applications and conditioning analysis showing polynomial fitting with different degrees, residual vs conditioning trade-offs, rank-deficient matrix analysis via SVD, and regularization effects using the L-curve method. These examples demonstrate practical applications of least squares methods and the importance of conditioning in numerical stability.

6 Applications and Special Cases

6.1 Polynomial Fitting

One of the most common applications of least squares is polynomial fitting to data.

Example 6.1 (Polynomial Least Squares)**.** *Given data points (t_i, y_i) for $i = 1, \ldots, m$, fit a polynomial of degree n:*

$$p(t) = c_0 + c_1 t + c_2 t^2 + \cdots + c_n t^n$$

This leads to the Vandermonde system:

$$\begin{pmatrix} 1 & t_1 & t_1^2 & \cdots & t_1^n \\ 1 & t_2 & t_2^2 & \cdots & t_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & t_m & t_m^2 & \cdots & t_m^n \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}$$

Vandermonde matrices become increasingly ill-conditioned as the polynomial degree increases, making QR decomposition essential for numerical stability.

6.2 Rank-Deficient Problems

When A does not have full column rank, the least squares problem has infinitely many solutions.

Definition. Minimum Norm Solution

Among all solutions to the least squares problem, the minimum norm solution is:

$$x^+ = A^+ b$$

where A^+ is the Moore-Penrose pseudoinverse.

The pseudoinverse can be computed using the SVD: if $A = U\Sigma V^T$, then:

$$A^+ = V\Sigma^+ U^T$$

where Σ^+ is formed by taking the reciprocal of nonzero singular values.

6.3 Regularization

For ill-conditioned problems, regularization can improve solution quality.

Definition. Ridge Regression

The ridge regression problem is:

$$\min_x \|Ax - b\|_2^2 + \lambda \|x\|_2^2$$

where $\lambda > 0$ is the regularization parameter.

This is equivalent to solving the augmented least squares problem:

$$\min_x \left\| \begin{pmatrix} A \\ \sqrt{\lambda} I \end{pmatrix} x - \begin{pmatrix} b \\ 0 \end{pmatrix} \right\|_2^2$$

The L-curve method provides a systematic way to choose the regularization parameter by plotting $\|Ax - b\|_2$ versus $\|x\|_2$ for different values of λ.

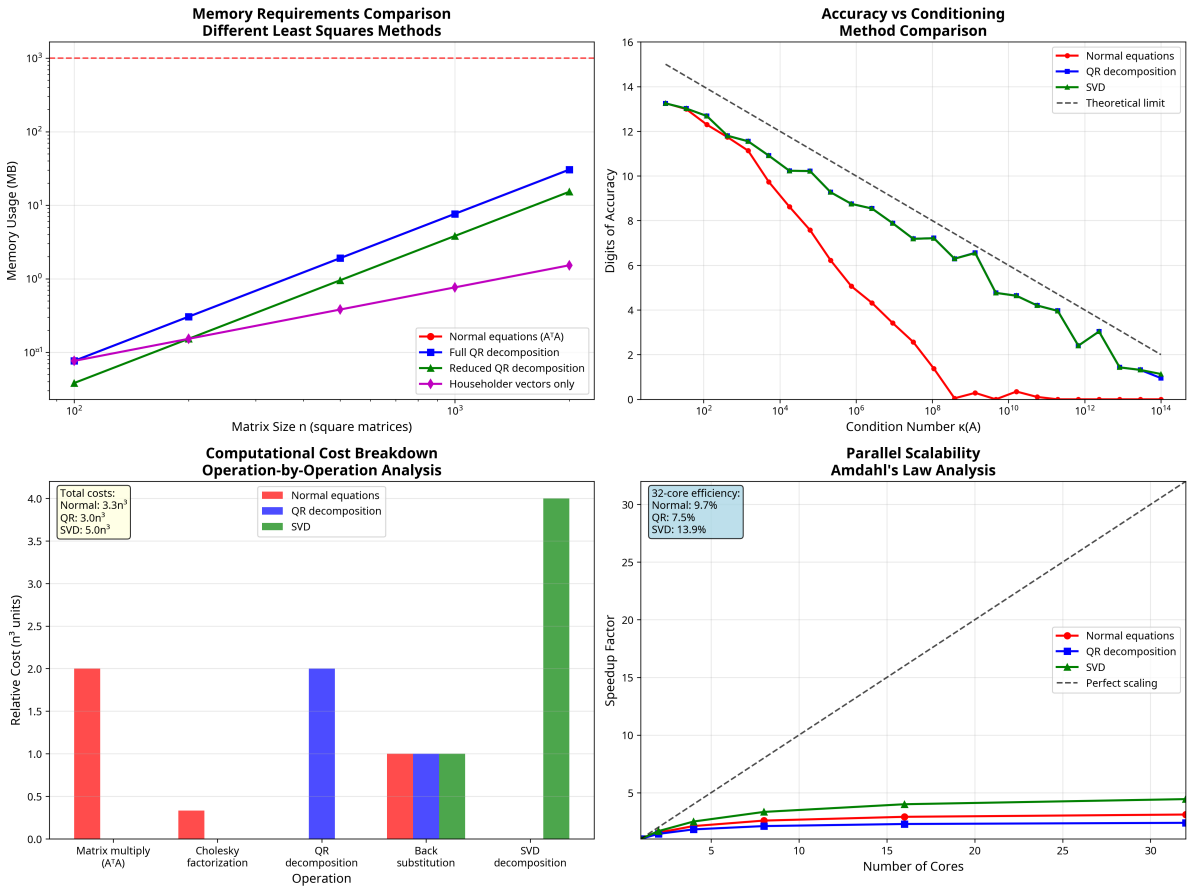

Figure 4: Computational methods comparison analyzing memory requirements, accuracy vs conditioning, cost breakdown by operation, and parallel scalability. This comprehensive analysis helps in selecting the appropriate method based on problem characteristics and computational constraints.

7 Computational Considerations

7.1 Memory Requirements

Different methods have varying memory requirements:

- **Normal equations:** Require storing $A^T A$ ($O(n^2)$ memory)
- **Full QR:** Require storing both Q and R ($O(mn)$ memory)
- **Householder vectors:** Store only the Householder vectors ($O(mn)$ memory, implicit Q)

7.2 Parallel Implementation

QR decomposition algorithms have different parallel characteristics:

- **Normal equations:** Matrix multiplication $A^T A$ is highly parallel
- **Householder QR:** Sequential dependency in column elimination
- **Block algorithms:** Enable better parallelization through blocking

7.3 When to Use Each Method

- **Normal equations:** Use when A is well-conditioned ($\kappa(A) \lesssim 10^6$) and multiple right-hand sides exist
- **QR decomposition:** Use for general-purpose robust solving, especially when numerical stability is critical
- **SVD:** Use for rank-deficient problems or when maximum numerical stability is required

Practice Problems

1. Implement both normal equations and QR-based solutions for a polynomial fitting problem. Compare their accuracy for different polynomial degrees and noise levels.
2. Create an ill-conditioned least squares problem and demonstrate the stability difference between normal equations and QR decomposition.
3. Implement the L-curve method for choosing the regularization parameter in ridge regression.
4. Compare the computational cost and memory usage of different QR algorithms (Gram-Schmidt, Modified Gram-Schmidt, Householder) for matrices of varying sizes.
5. Analyze the effect of condition number on solution accuracy for both normal equations and QR decomposition methods.